

Eshaan Gupta

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SUMMARY

AI/ML engineer specializing in computer vision and deep learning, with hands-on experience across research, industry, and deployment. Published at **ICMLSC'26**. Focused on building models that work — not just in notebooks, but in production.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

University School of Automation and Robotics

Guru Gobind Singh Indraprastha University, Delhi

CGPA: 8.9

Relevant Coursework: Machine Learning, Computer Vision, Deep Learning, Reinforcement Learning, Digital Image Processing

Class XII & X

Mount St Mary's School, Delhi

Percentage: 94% (XII), 92% (X)

PUBLICATIONS

- **DR VisionNet: A vision network for classification of diabetic retinopathy**, 2026 *International Conference on Machine Learning and Soft Computing (ICMLSC'26)*, Osaka, Japan.
 - Proposed a novel Deep CNN architecture and specialized preprocessing strategies for Diabetic Retinopathy classification, achieving 97% accuracy on the Messidor-2 dataset and outperforming state-of-the-art pre-trained models.

EXPERIENCE

Terafac Technologies Pvt Ltd

Jr. Machine Learning and Data Engineer

Feb 2026 – Present

- Built an object segmentation & tracking pipeline with **96% mask accuracy** using Albumentations augmentation.
- Explored **Neural Radiance Fields (NeRF)** for 3D scene reconstruction and novel view synthesis.

Quamin Tech Solutions LLP

Full Stack Web Developer

Jul 2024 – Mar 2025

- Developed and maintained web applications using the MERN stack (MongoDB, Express.js, React, Node.js).
- Integrated an NLP-based chatbot and market analyzer into the farmer dashboard, boosting user engagement by 20%.

Reliance Power Plant Ltd.

Machine Learning Intern

Jul 2024 – Aug 2024

- Implemented an LSTM-based grid frequency prediction model, achieving 80% accuracy.
- Conducted exploratory data analysis on power generation datasets, identifying key efficiency patterns.

PROJECTS

Diabetic Retinopathy Detection

[GitHub Repository Link](#)

- Engineered a multi-class classification pipeline for detecting Diabetic Retinopathy from retinal fundus images using deep learning models (InceptionV3, EfficientNetV2B0, ResNet18/34/50, ConvNext Small, custom CNN). Achieved 97% accuracy across 5 severity classes, with preprocessing improving model consistency by 3% compared to raw data.

Swastha Health

[Live Application Link](#)

- Developed a health monitoring web app used by over 10 individuals to upload medical reports, extract vital signs, and track trends. Secured with hashed passwords, powered by MongoDB, and hosted on Render (backend) and Netlify (frontend), with Docker containerization managed via DockerHub.

TECHNICAL SKILLS

- **AI & Computer Vision:** PyTorch, TensorFlow, OpenCV, YOLO, RF DETR, Image Segmentation, Object Detection, NeRF.
- **Deep Learning Architectures:** CNNs, RNNs, LSTMs, GANs, Reinforcement Learning, Time Series Forecasting.
- **Medical Imaging & Research:** Fundus Image Analysis, Multi-class Classification, Benchmark Evaluation, Novel Architecture Design.
- **MLOps & Deployment:** Docker, FastAPI/Flask, REST APIs, Git, Linux, Google Colab.
- **Programming & Data:** Python, SQL, EDA, Jupyter.
- **Full-Stack:** MERN Stack, Node.js, React, MongoDB.

HONOURS

- Brij Gala Goel Award for Topper in Mathematics
- 3rd Position, SRM Hackathon
- Top 9, Great Bengaluru Hackathon
- Top 30, Smart India Hackathon, 2023 (Team Leader)
- Top 30, Smart India Hackathon, 2024
- Top 5, The Annual Flagship Hackathon, SGGSCC, 2024